KRISHI MITRA – TECHNICAL AUDIT & AI/ML ACCURACY REPORT

Comprehensive Architecture Verification, Git History Forensics, Accuracy Audit & Viva Defense Dossier

PROJECT: Krishi Mitra – AI-Based Farmer Query Support & Decision Assistance	**DEGREE / BRANCH:** B.Tech – Computer Science & Engineering
INSTITUTION: United Institute of Technology	**FACULTY GUIDE:** Dr. Mohd. Osama (PhD)
PROJECT TEAM: Amol Singh, Ashish Kumar, Amit Kumar, Shyam Babu	**AUDIT DATE & REPO HEAD:** Oct 2026 | Commit: `e4ff111`

1. Executive Summary

This independent technical audit provides a rigorous, read-only analysis of the machine learning architectures, disease prediction pipeline, dataset distributions, and claimed accuracy metrics in the **Krishi Mitra** codebase. The system has evolved from a simple sequential convolutional neural network into a sophisticated **4-model ensemble** equipped with Test-Time Augmentation (TTA), a 4,648-dimensional deep botanical verifier, and real-time agricultural APIs.

■ CRITICAL AUDIT FINDINGS

1. Architecture Shift: Transitioned from a single 3-block CNN (134 MB) to an ensemble of **MobileNetV2** backbones plus the baseline CNN. Neither ResNet nor EfficientNet is present in the codebase.

2. Accuracy Status: The commit message `e4ff111` claims *'95% acc'* and UI badges show *'92% Accuracy'*. However, **no evaluation logs, CSVs, or confusion matrix dumps exist on disk**. Under strict read-only audit rules, computing ~19,000 forward passes on 3,203 test images was omitted. The 95% figure remains *unverified developer assertion*.

3. Leaf Verifier Gate: An advanced 4-stage botanical & deep representation verification filter (`verify_is_tomato_leaf`) effectively rejects non-tomato foliage and non-leaf uploads.

4. Confidence Calibration: Production inference applies an artificial post-processing floor forcing any top-class prediction with <91.2% confidence up to **91.2% – 98.6%**.

5. Missing & Duplicate Artifacts: `unet_leaf_segmentation.h5` is missing from disk (causing graceful fallback to OpenCV contours). `plant_disease_model.h5` and `...mobilenet_candidate.h5` are bit-for-bit identical duplicates (SHA-256 matched).

2. Current Model Architecture & System Identification

The production Django application (`disease_service.py`) employs a multi-model inference pipeline comprising four deep neural networks, a binary linear support verifier, and a 4-stage botanical gate:

Identifier	File Name	Architecture Type	Input Shape	Parameter & Layer Composition	Size
m_main	`plant_disease_model.h5`	MobileNetV2 (Functional)	(224, 224, 3)	7 Layers: MobileNetV2 Backbone + GAP (1280) + BatchNorm + Dense (256, ReLU) + Dropout (0.4) + Dense (10)	23.08 MB
m_mobilenet	`plant_disease_model_mobilenet_candidate.h5`	MobileNetV2 (Functional)	(224, 224, 3)	Identical bit-for-bit duplicate of `m_main` (identical SHA-256 hash).	23.08 MB
m_focal	`plant_disease_model_focal_candidate.h5`	MobileNetV2 (Functional)	(224, 224, 3)	7 Layers: MobileNetV2 Backbone + GAP (1280) + BatchNorm + Dense (256, ReLU) + Dropout (0.4) + Dense (10)	13.42 MB
m_seg	`plant_disease_model_segmented_candidate.h5`	MobileNetV2 (Functional)	(224, 224, 3)	7 Layers: MobileNetV2 Backbone + GAP (1280) + BatchNorm + Dense (256, ReLU) + Dropout (0.4) + Dense (10)	13.42 MB
m_orig	`plant_disease_model_original_backup.h5`	3-Block CNN (Sequential)	(224, 224, 3)	11 Layers: Conv2D(32) + MaxPool + Conv2D(64) + MaxPool + Conv2D(128) + MaxPool + Flatten(100k) + Dense(128) + Dropout(0.5) + Dense(10)	134.09 MB
verifier	`tomato_leaf_verifier.npz`	LogReg + Exemplar Bank	(4648,)	Linear classifier weights (4648-D) + 76 positive and 120 negative non-tomato deep feature exemplars.	2.38 MB

unet	`unet_leaf_segmentation.h5`	Supervised U-Net	(256, 256, 3)	**Missing from disk**. Pipeline handles missing file gracefully; falls back to OpenCV contour segmentation.	0 B (Missing)

4-Model Ensemble Fusion Formula:

In `disease_service.py:L424`, predictions are fused using weighted linear pooling:

P_ensemble = 0.48 · p_focal + 0.30 · p_main + 0.14 · p_orig + 0.08 · p_seg

(Note: In the offline evaluation script `check_accuracy.py`*, a 3-way blend is used:* ***0.50 · p_focal + 0.35 · p_main + 0.15 · p_orig****, excluding m_seg because test data lacks segmented masks).*

Test-Time Augmentation (TTA) Views:

- **View 1:** Original RGB image (224x224, normalized [0, 1]).
- **View 2:** Horizontal Flip (`np.fliplr`).
- **View 3:** Vertical Flip (`np.flipud`).
- **View 4:** 5% Center Zoom Crop (crops outer 5% border, resized to 224x224).
- **View 5:** LAB CLAHE View (L-channel histogram equalization with clipLimit=2.0 to sharpen subtle fungal lesions).
- **Segmented Views:** Background-masked leaf ROI + horizontal flip (2 views passed to `m_seg`).

4-Stage Botanical & Species Verifier Gate (verify_is_tomato_leaf):

1. **Stage 1 (Filename Guard):** Rejects 72+ non-tomato crop keywords (rice, wheat, apple, potato, selfie, aadhaar, etc.).
2. **Stage 2 (Spectral & ExG Filter):** Evaluates green chlorophyll coverage (HSV), necrotic lesions, and Excess Green Index (ExG = 2G - R - B > 12.0). Rejects non-foliage photos if green ratio < 2.5% or ExG ratio < 1.5%.
3. **Stage 3 (Morphology Filter):** Rejects narrow strap-like monocot leaves (Rice, Wheat, Grass) if aspect ratio > 3.25.
4. **Stage 4 (4,648-D Logit Verifier):** Extracts GAP and Dense feature vectors across all models, evaluates logistic regression score and nearest-neighbor cosine similarity against 76 positive tomato and 120 negative non-tomato exemplars.

3. Current Model Accuracy & Evidence Audit

To satisfy rigorous academic audit standards, all recorded artifacts, commits, logs, and datasets were inspected. The table below details reported vs. scientifically verifiable metrics:

Metric	Claimed / Displayed Value	Verification Status	Forensic Source & Evidence Location
Ensemble Test Accuracy	95.0%	**Not verified from available evidence**	Stated in Git commit message `e4ff111`. No output log or evaluation file exists.
CNN Model Accuracy	92% (was 98%)	**Unverified UI counter**	Frontend counter in `templates/home.html:L108`. Working tree changed 98 to 92.
Inference Confidence	91.2% – 99.8%	**Post-processing floor**	`disease_service.py:L441-L451` artificially clamps confidence to ≥ 91.2%.
Validation Accuracy	Not reported	**Not available**	No separate validation split kept in `split_dataset.py`; test set was used during training.
Precision / Recall / F1	Not reported	**Not verified**	No saved `classification_report` text or JSON export exists in repository.
Confusion Matrix	Not reported	**Not verified**	No saved confusion matrix dump exists in repository.

Exact Dataset Image Distribution (PlantVillage Tomato Subset):

Verified from `research_results_export/dataset/dataset_distribution.csv`. Split ratio: 80% Train, 20% Test.

Class Name	Train Count (80%)	Test Count (20%)	Total Images	Class Share
Tomato_Bacterial_spot	1,701	426	2,127	13.3%
Tomato_Early_blight	800	200	1,000	6.3%
Tomato_Late_blight	1,522	379	1,901	11.9%
Tomato_Leaf_Mold	761	191	952	6.0%
Tomato_Septoria_leaf_spot	1,416	355	1,771	11.1%
Tomato_Spider_mites_Two_spotted_spider_mite	1,340	336	1,676	10.5%
Tomato__Target_Spot	1,123	281	1,404	8.8%

Tomato__Tomato_YellowLeaf__Curl_Virus	2,566	642	3,208	20.1%
Tomato__Tomato_mosaic_virus	298	75	373	2.3%
Tomato_healthy	1,267	318	1,585	9.9%
TOTALS	**12,794**	**3,203**	**15,997**	**100.0%**

4. Previous Model Architecture & Baseline Accuracy

Prior to commit `e4ff111`, Krishi Mitra utilized a single, custom Sequential 3-Block CNN:
• **Architecture:** Conv2D(32, 3x3) → MaxPool → Conv2D(64, 3x3) → MaxPool → Conv2D(128, 3x3) → MaxPool → Flatten (100,352) → Dense(128, ReLU) → Dropout(0.5) → Dense(10, Softmax).
• **Model Weight File:** Preserved on disk as `plant_disease_model_original_backup.h5` (134,090,056 bytes).
• **Training Parameters:** 10 Epochs, Batch Size: 32, Adam Optimizer, Categorical Crossentropy, 224x224 input resolution.
• **Baseline Test Accuracy:** Not verified from available evidence. The UI previously displayed 98% (or 92%), but no evaluation log was preserved.
• **Limitations of Baseline:** Large parameter size (~12.9M params, 134 MB), susceptible to background noise, lacks multi-scale feature representation.

5. Accuracy Improvement Techniques Audit

Technique	Status	Implementation Location	How It Operates in Krishi Mitra	Empirical Proof?
Transfer Learning	**Yes**	`disease_service.py:L46`	Replaces from-scratch CNN with ImageNet-pretrained MobileNetV2 backbone.	Not verified
MobileNetV2 Backbone	**Yes**	`plant_disease_model.h5`	Depthwise separable convolutions reduce params while maintaining feature richness.	Not verified
Fine-Tuning Head	**Yes**	`H5 Layer Configuration`	Custom head: GAP (1280) + BatchNorm + Dense(256) + Dropout(0.4) + Dense(10).	Not verified
Data Augmentation	**Yes**	`train_model.py:L12`	Rotation 20 deg, zoom 0.2, horizontal flip during training.	Not verified
Image Preprocessing & CLAHE	**Yes**	`disease_service.py:L220`	Normalized [0, 1] + Contrast Limited Adaptive Histogram Equalization on L-channel.	Not verified
Class Balancing / Weights	**No**	`train_model.py`	class_weight is not passed to model.fit(); unweighted training.	No
Focal Loss	**Candidate Name Only**	`...focal_candidate.h5`	Model candidate filename contains 'focal', but no focal loss function is in code.	Not verified
Label Smoothing	**No**	`Entire Codebase`	Not implemented in any training or loss function.	No
Learning Rate Scheduling	**No**	`train_model.py`	Uses static Adam learning rate (0.001); no decay scheduler callback.	No
Dropout & Regularization	**Yes**	`H5 Layer Configuration`	Dropout(0.5) in 3-block CNN; Dropout(0.4) & BatchNorm in MobileNetV2 models.	Not verified
Ensemble Learning	**Yes**	`disease_service.py:L416`	Blends 4 models (3 MobileNetV2 variants + 1 3-Block CNN).	Not verified
Weighted Soft Voting	**Yes**	`disease_service.py:L424`	Weights: 0.48 (focal) + 0.30 (main) + 0.14 (orig) + 0.08 (seg).	Not verified
Test-Time Augmentation (TTA)	**Yes**	`disease_service.py:L393`	5-view TTA: Original, H-Flip, V-Flip, 5% Center Zoom Crop, LAB CLAHE.	Not verified
Botanical & Species Gate	**Yes**	`disease_service.py:L254`	4-Stage Gate: Keywords + HSV/ExG spectral filter + aspect ratio + 4648-D LogReg verifier.	Not verified
Confidence Clamping Floor	**Yes**	`disease_service.py:L441`	Post-processing logic forces confidence to min(98.6, 91.2 + raw_top*7.5) if below 91.2%.	Artificially forces ≥91.2%

6. Model Comparison Table & Scientific Audit Answers

Model / System	Version	Architecture	Dataset Split	Classes	Test Acc. (%)	Techniques Added	Scientific Status

Original CNN	Previous (`04a806c`)	3-Block CNN (Conv2D 32-64-128, Dense 128)	PlantVillage Tomato (80/20: 3,203 test)	10	*Not verified* (UI displayed 98% / 92%)	Augmentation, Dropout (0.5), Adam	Baseline Model (Weights preserved in backup file)
MobileNetV2 Main	Current Candidate	MobileNetV2 + GAP + BatchNorm + Dense 256	PlantVillage Tomato (80/20)	10	*Not verified*	Transfer learning, MobileNetV2, BatchNorm, Dropout (0.4)	Duplicate of mobilenet_candidate.h5 (identical SHA-256)
MobileNetV2 Focal	Current Candidate	MobileNetV2 + GAP + BatchNorm + Dense 256	PlantVillage Tomato (80/20)	10	*Not verified*	Transfer learning, MobileNetV2, BatchNorm, Dropout (0.4)	Directly comparable architecture
MobileNetV2 Segmented	Current Candidate	MobileNetV2 + GAP + BatchNorm + Dense 256	PlantVillage Tomato (Masked ROI)	10	*Not verified*	Background removal, Transfer learning, MobileNetV2	Trained on masked leaf ROIs
4-Model Ensemble + TTA	Current (`e4ff111`)	Ensemble (3x MobileNetV2 + 1x 3-Block CNN)	PlantVillage Tomato (80/20: 3,203 test)	10	*Not verified* (Commit claims 95%)	4-Model Soft Voting, 5-View TTA, CLAHE, Verifier Gate	Architecturally superior; empirical 95% unverified

Answers to Critical Scientific Questions:

- **What was the previous verified test accuracy?** Not verified from available evidence. UI displays showed 98% or 92%, but no log file or evaluation report was ever preserved.
- **What is the current verified test accuracy?** Not verified from available evidence. Commit message `e4ff111` asserts 95%, but no empirical evaluation log exists.
- **How many percentage points has accuracy changed?** Cannot be certified without running the evaluation script across the test set.
- **Is the improvement experimentally proven?** No. The architectural improvements are fully coded and functional, but lack saved empirical test set documentation.
- **Were both models evaluated on the same dataset split?** Yes. Both target the 10-class PlantVillage Tomato subset (3,203 test images) with identical class indices.
- **Could data leakage explain apparent improvements?** During original CNN training, `validation_data=test_data` was passed directly. Furthermore, production inference clamps confidence to ≥91.2%, which must not be confused with true test accuracy.

7. Complete Technology & Tool Inventory

Category	Technology / Library	Version	Project Role & Purpose	Active Status & Notes
A. Languages	Python JavaScript (ES6+) HTML5 / CSS3	3.11.9 Standard Standard	Backend logic, deep learning, web views, APIs. Speech recognition, audio playback, theme toggle. Responsive UI, templates, and styling.	Actively Used. Actively Used. Actively Used.
B. Backend	Django python-decouple whitenoise	5.2.6 3.8 6.6.0	Full-stack MVC web framework & ORM. Environment decoupling in settings.py. Static file server (installed; middleware not declared).	Actively Used. Actively Used. Declared in requirements.txt.
C. Frontend	Bootstrap AOS Google Fonts	5.3.3 2.3.4 Online	Responsive styling, grid, modals, navbar. Animate-on-scroll CSS effects. Inter & Outfit typography.	Actively Used (CDN). Actively Used (CDN). Actively Used (CDN).
D. AI / ML	TensorFlow Keras MobileNetV2 Custom CNN U-Net	2.21.0 3.13.2 1.00 224 Custom Custom	Deep learning execution engine. High-level model loading and layer extraction. Pretrained transfer learning backbones. Legacy 3-block CNN baseline. Supervised leaf segmentation architecture.	Actively Used (Not in reqs). Actively Used. Actively Used in ensemble. Actively Used in ensemble. Weights missing from disk.
E. Computer Vision	OpenCV (opencv-python) Pillow (PIL)	4.13.0.92 10.0.1	HSV/LAB filtering, CLAHE, contours, morphology. Image loading, byte streams, cropping.	Actively Used (Not in reqs). Actively Used (In reqs).

F. Data Science	NumPy scikit-learn SciPy h5py Matplotlib	2.4.3 1.8.0 1.17.1 3.14.0 3.11.1	Array math, feature vector stacking, scaling. Evaluation metrics (classification_report, matrix). Scientific algorithms. HDF5 model weights reader. Loss/accuracy curve plotting.	Actively Used (Not in reqs). Used in evaluation scripts. Installed dependency. Actively Used by TF. Used in offline scripts.
G. Database	SQLite3	Embedded	Relational database storing scans, contacts, advisory.	Actively Used (db.sqlite3).
H. APIs & Web	Open-Meteo API Agmarknet 2.0 API Wikipedia API requests	Public Public Public 2.32.3	Real-time weather forecast & rain probability. Live APMC wholesale market prices. Fallback agronomy knowledge retrieval. HTTP client for REST calls.	Live & Connected (No key). Live & Connected (No key). Live & Connected (No key). Actively Used (In reqs).
I. Voice & Trans.	Web Speech Recognition Web Speech Synthesis Google Translate Widget core/translation.py	Browser API Browser API CDN 0 Bytes	Voice speech-to-text (hi-IN, en-IN). Voice text-to-speech spoken answers. Client-side multilingual translation (8 languages). Empty Python module placeholder.	Actively Used. Actively Used. Actively Used. Unused (Empty).
J. Doc Generation	python-docx ReportLab	1.2.0 5.0.0	DOCX document export. PDF report generation canvas.	Used in offline generators. Used in report generation.

8. Complete Project Feature Audit

Feature Module	Implementation Classification	API / Service Dependency	Technical Mechanism & Status
Plant Disease Detection	**Implemented in code**	Internal Deep Learning	4-Model Ensemble with 5-View TTA, temperature sharpening, and visual overlay. Fully functional.
Botanical Leaf Validation	**Implemented in code**	Internal CV Gate	4-Stage Gate: Filename keywords, HSV/ExG spectral filter, aspect ratio, and 4648-D LogReg verifier.
Supervised U-Net Lesion Seg.	**Partially Implemented**	Internal U-Net Model	Code fully implemented in segmentation_service.py, but weights file missing; falls back to OpenCV.
Live Agmarknet Mandi Prices	**Implemented & Connected**	api.agmarknet.gov.in/v1	Queries live GoI Agmarknet dashboard for 36 States/UTs. Includes offline directory fallback.
Live Weather & Farm Advisory	**Implemented & Connected**	Open-Meteo Weather API	Fetches live temp, rain prob, humidity for 100+ Indian districts; generates dynamic farming advice.
Crop Advisory & Soil NPK	**Implemented in code**	Internal Rule Engine	Evaluates soil N-P-K & pH against ICAR agronomic thresholds; calculates exact Urea, DAP, MOP dosages.
Government Schemes Portal	**Implemented in code**	Internal Informational	Bilingual directory covering PM-KISAN, PMFBY, KCC, Soil Card, PMKSY with application links.
Multilingual Voice Assistant	**Implemented & Connected**	Web Speech + APIs	Browser Web Speech API (STT & TTS) connected to Django assistant_service (Weather, Mandi, Wikipedia).
Database Logging & Models	**Implemented & Active**	SQLite (db.sqlite3)	3 ORM models (DiseaseDetectionLog, ContactMessage, CropAdvisoryLog) actively recording scans.

9. Identified Errors, Technical Risks & Anomalies

- **Missing U-Net Weights File:** segmentation_config.py:L27 expects 'unet_leaf_segmentation.h5', which is missing from disk. True deep lesion segmentation is dormant and falls back to OpenCV contours.
- **Duplicate Model Weights File:** plant_disease_model.h5 and plant_disease_model_mobilenet_candidate.h5 share the exact identical SHA-256 hash (96CD4214B1FCCAC77DBCE300EE2AF2AF14933B02299A4A77DE795079FB497465), wasting 23 MB of disk space.
- **Incomplete requirements.txt:** Essential runtime libraries (TensorFlow, OpenCV, scikit-learn, NumPy, SciPy, h5py, Matplotlib) are installed in the venv but missing from requirements.txt, causing setup failures on clean environments.

- **Artificial Confidence Inflation:** disease_service.py:L441-L451 explicitly forces confidence to min(98.6, 91.2 + raw_top*7.5) whenever confidence < 91.2%. This gives a false illusion of certainty to users.
- **Discrepancy Between Inference & Evaluation:** Production uses 4 models + 5 TTA views, whereas check_accuracy.py uses 3 models + 2 TTA views and excludes m_seg. The evaluation script does not mirror production.
- **Unused API Key & Default Secret:** settings.py defines AGMARKNET_API_KEY, but mandi_service.py never uses it (queries public endpoint). Additionally, SECRET_KEY uses an insecure default string.
- **Missing Training Code for MobileNetV2 Models:** train_model.py only trains the old 3-block CNN. The training scripts, focal loss code, and training histories for the current MobileNetV2 models are absent.

10. Final Conclusions & Concise Viva Defense Summary

■ **CONCISE DIRECT VIVA ANSWERS**

1. Which model did we use before?
A single, custom Sequential **3-Block Convolutional Neural Network** (Conv2D: 32 → 64 → 128 → Dense: 128 → Dropout: 0.5 → Dense: 10), preserved in `plant_disease_model_original_backup.h5` (134 MB).

2. Which model or models are we using now?
A **4-Model Ensemble** combining 3 transfer-learning **MobileNetV2** variants (`m_focal`, `m_main`, `m_seg`) and the legacy **3-Block CNN** (`m_orig`), plus a 4,648-D Deep Feature LogReg Verifier Gate (`tomato_leaf_verifier.npz`).

3. What techniques were added to improve accuracy?
Transfer learning (MobileNetV2 backbones), multi-model weighted soft voting (0.48, 0.30, 0.14, 0.08), 5-view Test-Time Augmentation (TTA), LAB CLAHE histogram equalization, temperature calibration, and a 4-stage botanical & deep feature verification gate.

4. What was the previous verified test accuracy?
Not verified from available evidence. UI counters displayed 98% or 92%, but no evaluation log was committed or preserved.

5. What is the current verified test accuracy?
Not verified from available evidence. The commit message mentions '95% acc', but no evaluation log, confusion matrix, or classification report exists in the repository.

6. What is the actual measured improvement?
Not verified from available evidence. The architectural improvements are fully implemented in code, but empirical before-and-after percentage gains cannot be certified without running the evaluation script across the test set.